

Software Programming

IGS1931

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Functions Practice

Software Programming (IGS1931) · Lecture

In-class exercises with Python functions and common bugs.

Today's practice

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- Call one function from another function.
- Read an error message and find the line that caused it.
- Check the output when code runs but gives the wrong answer.
- Write small functions with parameters, defaults, and return values.

Concept

page3

A function can call another function you've already written. In the next exercise, one function calculates an area and another uses that result to calculate the cost.

Question: rectangle_area and rectangle_cost

page4

QUESTION

Define `rectangle_area(length, width)` that returns `length width`. Then define `rectangle_cost(length, width, price_per_sqm)` that calls `rectangle_area` internally to get the area, then multiplies by the price per square meter.

```
rectangle_cost(4, 5, 20000) returns 400000  
rectangle_cost(3, 3, 45000) returns 405000
```

Answer: rectangle_area and rectangle_cost

page5

ANSWER

Code

```
def rectangle_area(length, width):  
    return length * width  
  
def rectangle_cost(length, width, price_per_sqm):  
    area = rectangle_area(length, width)  
    return area * price_per_sqm  
  
print(rectangle_cost(4, 5, 20000))  
print(rectangle_cost(3, 3, 45000))
```

Output

400000 405000

Functions recap

page6

- Composition: one function calls another.
- Parameters are the names in def. Arguments are the values in a call.
- A default value lets you leave out an argument, as in `def f(x, y=10)`.
- `return` sends a value back to the caller. `print` only displays it.
- Without a return statement, a function returns `None`.

Common Python errors

page7

Five raise an error. The last one runs but gives the wrong answer.

SyntaxError

```
def f(x)
```

Missing colon

NameError

```
print(missing_value)
```

The name was never defined

ZeroDivisionError

```
10 / 0
```

Cannot divide by zero

IndentationError

```
if True:  
print("Hi")
```

The body needs an indent

TypeError

```
"5" + 2
```

A string and an integer cannot be added

Logic error

```
average = (2 + 4) / 3
```

Runs, but the average should be 3.0

Debug 1: square

page8

QUESTION

Expected: square(4) should return 16.

```
def square(number):  
    print(number ** 2)  
  
print(square(4))
```

What appears on screen? Why are there two lines?

Debug 1: square

page9

ANSWER

In the starter code, square prints 16 but returns None. The outer print then displays None.

```
def square(number):  
    return number ** 2  
  
print(square(4))
```

Output

16

Debug 2: add_tax

page10

QUESTION

Expected: `add_tax(100, 10)` should return 110.0 (price plus 10% tax).

```
def add_tax(pricee, tax_rate):  
    return price + price * tax_rate / 100  
  
print(add_tax(100, 10))
```

Run the code. What name does the error mention?

Debug 2: add_tax

page11

ANSWER

The starter code defines pricee but uses price. Python cannot find price.

```
def add_tax(price, tax_rate):  
    return price + price * tax_rate / 100  
  
print(add_tax(100, 10))
```

Output

110.0

Debug 3: greet

page12

QUESTION

Expected: `greet("Aziz")` returns "Hello, Aziz!", `greet("Dilnoza", "Hi")` returns "Hi, Dilnoza!".

```
def greet(greeting="Hello", name):  
    return f"{greeting}, {name}!"  
  
print(greet("Aziz"))  
print(greet("Dilnoza", "Hi"))
```

Python stops at the `def` line. Check the order of the parameters.

Debug 3: greet

page13

ANSWER

The starter code puts greeting before name. Parameters without defaults must come first.

```
def greet(name, greeting="Hello"):
    return f"{greeting}, {name}!"

print(greet("Aziz"))
print(greet("Dilnoza", "Hi"))
```

Output

Hello, Aziz! Hi, Dilnoza!

Debug 4: perimeter

page14

QUESTION

Expected: `perimeter(4, 3)` returns 14, `perimeter(10, 5)` returns 30.

```
def perimeter(length, width):  
    return length + width  
  
print(perimeter(4, 3))  
print(perimeter(10, 5))
```

The code runs, but the answers are wrong. Compare them with the expected values.

Debug 4: perimeter

page15

ANSWER

The starter code adds length and width once. A perimeter needs twice that sum.

```
def perimeter(length, width):  
    return 2 * (length + width)  
  
print(perimeter(4, 3))  
print(perimeter(10, 5))
```

Output

14 30

Question 1: tip_amount

page16

QUESTION

Write `tip_amount(bill, tip_percent=15)` returning `bill tip_percent / 100`. Default tip: 15%.

```
tip_amount(80) returns 12.0  
tip_amount(50, 20) returns 10.0
```

Answer 1: tip_amount

page17

ANSWER

Code

```
def tip_amount(bill, tip_percent=15):  
    return bill * tip_percent / 100  
  
print(tip_amount(80))  
print(tip_amount(50, 20))
```

Output

12.0 10.0

Question 2: speed

page18

QUESTION

Write `speed(distance, time)` returning `distance / time`.

`speed(150, 3)` returns `50.0`
`speed(80, 2)` returns `40.0`

Answer 2: speed

page19

ANSWER

Code

```
def speed(distance, time):  
    return distance / time  
  
print(speed(150, 3))  
print(speed(80, 2))
```

Output

50.0 40.0

Question 3: total_with_tax

page20

QUESTION

Write `total_with_tax(price, tax_rate)` returning $\text{price} + \text{price} \times \text{tax_rate} / 100$.

```
total_with_tax(100, 8) returns 108.0  
total_with_tax(250, 12) returns 280.0
```

Answer 3: total_with_tax

page21

ANSWER

Code

```
def total_with_tax(price, tax_rate):  
    return price + price * tax_rate / 100  
  
print(total_with_tax(100, 8))  
print(total_with_tax(250, 12))
```

Output

108.0 280.0

Question 4: free_fall_velocity

page22

QUESTION

Write `free_fall_velocity(time, g=9.8)` returning `g time`. Default to Earth's gravity, like `free_fall_distance` in Week 2.

`free_fall_velocity(2)` returns 19.6

`free_fall_velocity(3, 1.62)` returns 4.86 (Moon's gravity)

Answer 4: free_fall_velocity

page23

ANSWER

Code

```
def free_fall_velocity(time, g=9.8):  
    return g * time  
  
print(free_fall_velocity(2))  
print(free_fall_velocity(3, 1.62))
```

Output

19.6 4.86

Question 5: discounted_price

page24

QUESTION

Expected: `discounted_price(price, discount_percent)` should return $\text{price} - \text{price} \times \text{discount_percent} / 100$.

`discounted_price(200, 25)` returns 150.0
`discounted_price(80, 10)` returns 72.0

```
def discounted_price(price, discount_percent):  
    return price - price * discount_percent / 100  
  
print(discounted_price(200, 25))  
print(discounted_price(80, 10))
```

The code runs. Compare its output with the two values above.

Answer 5: discounted_price

page25

ANSWER

The starter code subtracts 25 rather than 25% of 200. It prints 175 and 70.

```
def discounted_price(price, discount_percent):  
    return price - price * discount_percent / 100  
  
print(discounted_price(200, 25))  
print(discounted_price(80, 10))
```

Output

150.0 72.0

Question 6: kinetic_energy

page26

QUESTION

Write `kinetic_energy(mass, velocity)` returning $0.5 \text{ mass velocity}^2$.

```
kinetic_energy(2, 3) returns 9.0  
kinetic_energy(10, 4) returns 80.0
```

Answer 6: kinetic_energy

page27

ANSWER

Code

```
def kinetic_energy(mass, velocity):  
    return 0.5 * mass * velocity ** 2  
  
print(kinetic_energy(2, 3))  
print(kinetic_energy(10, 4))
```

Output

9.0 80.0

Question 7: taxi_fare

page28

QUESTION

Write `taxi_fare(distance_km, rate_per_km=2000, base_fare=5000)` returning `base_fare + distance_km rate_per_km`.

`taxi_fare(5)` returns 15000
`taxi_fare(10, 2500)` returns 30000

Answer 7: taxi_fare

page29

ANSWER

Code

```
def taxi_fare(distance_km, rate_per_km=2000, base_fare=5000):  
    return base_fare + distance_km * rate_per_km  
  
print(taxi_fare(5))  
print(taxi_fare(10, 2500))
```

Output

15000 30000

Question 8: electricity_bill

page30

QUESTION

Write `electricity_bill(units_used, rate_per_unit=1200)` returning `units_used` `rate_per_unit`.

```
electricity_bill(150) returns 180000  
electricity_bill(200, 1500) returns 300000
```

Answer 8: electricity_bill

page31

ANSWER

Code

```
def electricity_bill(units_used, rate_per_unit=1200):  
    return units_used * rate_per_unit  
  
print(electricity_bill(150))  
print(electricity_bill(200, 1500))
```

Output

180000 300000

Question 9: split_bill_with_tip

page32

QUESTION

Expected: `split_bill_with_tip(total, num_people, tip_percent=10)` should add the tip to the total, then split evenly: $(\text{total} + \text{total tip_percent} / 100) / \text{num_people}$.

`split_bill_with_tip(100000, 4)` returns 27500.0
`split_bill_with_tip(60000, 3, 15)` returns 23000.0

```
def split_bill_with_tip(total, num_people, tip_percent=10):  
    tip = total * tip_percent / 100  
    return (total / num_people) + tip  
  
print(split_bill_with_tip(100000, 4))  
print(split_bill_with_tip(60000, 3, 15))
```

The tip is correct. What does the code divide by `num_people`?

Answer 9: split_bill_with_tip

page33

ANSWER

The starter code adds the full tip to each person's share. Divide the tipped total instead.

```
def split_bill_with_tip(total, num_people, tip_percent=10):  
    return (total + total * tip_percent / 100) / num_people  
  
print(split_bill_with_tip(100000, 4))  
print(split_bill_with_tip(60000, 3, 15))
```

Output

27500.0 23000.0

Question 10: average_speed

page34

QUESTION

Write `average_speed(distance1, time1, distance2, time2)` returning average speed over two legs of a trip: $(\text{distance1} + \text{distance2}) / (\text{time1} + \text{time2})$.

`average_speed(100, 2, 200, 2)` returns 75.0
`average_speed(50, 1, 100, 3)` returns 37.5

Answer 10: average_speed

page35

ANSWER

Code

```
def average_speed(distance1, time1, distance2, time2):  
    return (distance1 + distance2) / (time1 + time2)  
  
print(average_speed(100, 2, 200, 2))  
print(average_speed(50, 1, 100, 3))
```

Output

75.0 37.5

Question 11: paycheck_net

page36

QUESTION

Write `paycheck_net(gross_salary, tax_percent, insurance_flat)` returning take-home pay: start from `gross_salary`, subtract tax (`gross_salary tax_percent / 100`), then subtract the flat insurance deduction.

```
paycheck_net(5000000, 12, 150000) returns 4250000.0  
paycheck_net(3000000, 10, 100000) returns 2600000.0
```

Answer 11: paycheck_net

page37

ANSWER

Code

```
def paycheck_net(gross_salary, tax_percent, insurance_flat):  
    return gross_salary - (gross_salary * tax_percent / 100) -  
    insurance_flat  
  
print(paycheck_net(5000000, 12, 150000))  
print(paycheck_net(3000000, 10, 100000))
```

Output

4250000.0 2600000.0

What we practiced

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- One function can call another.
- An error message can point to the bug.
- Code can run without an error and still give the wrong answer.
- Parameters, defaults, and return values make functions useful in other code.

Homework and group projects

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Homework notebook

GitHub: https://github.com/abdukarimovhm/software-programming-lectures/blob/main/homework/week03_functions_homework_student.ipynb

Colab: https://colab.research.google.com/github/abdukarimovhm/software-programming-lectures/blob/main/homework/week03_functions_homework_student.ipynb

Group projects after the midterm

- Project work starts after the Week 9 briefing.
- Choose your team. The limit is five students.
- Design and build the project yourselves. Do not submit an AI-generated project.
- Possible topics: automation, encryption with dictionaries, games, research, or data visualization.
- Send your idea to abdukarimovh@inha.edu before Week 9 for feedback.
- Get approval from Prof. Kabir or the instructor before you start building.

Week 8: Midterm Week 9: Project briefing Week 14: Presentations